Limits – 3

Asymptotics of sequences

1 сентября 2026 года

Theoretical minimum.

Def. Knowing the limit is not enough: an olympiad problem usually asks for the rate. The standard answer has the form $x_n \sim Cn^{-\alpha}$ or $x_n = L + Cq^n + o(q^n)$, and the choice between the two is decided by the behaviour of f at the fixed point.

Theorem 1 (neutral fixed point). Let $x_n \to 0$, $x_n > 0$ and $x_{n+1} = x_n - ax_n^p + o(x_n^p)$ with $a > 0$, $p > 1$. Then

$$x_n \sim \Big(\frac{1}{a(p-1)n}\Big)^{1/(p-1)}.$$

In particular $x_{n+1} = x_n - x_n^2$ gives $nx_n \to 1$, while $x_{n+1} = \sin x_n$ (where $p = 3$, $a = \frac{1}{6}$) gives $x_n \sim \sqrt{3/n}$.

Theorem 2 (Stolz–Cesàro). If b_n is strictly increasing and unbounded and $\lim \frac{a_{n+1}-a_n}{b_{n+1}-b_n} = L$, then $\lim \frac{a_n}{b_n} = L$. This is the discrete analogue of l'Hôpital's rule, and it is exactly what proves Theorem 1: apply it to $y_n = x_n^{1-p}$.

Fact 1. The telescoping trick in powers of x_n: from $x_{n+1} = x_n - ax_n^p$ one gets $x_{n+1}^{1-p} - x_n^{1-p} \to a(p-1)$, and then Stolz finishes the job. This trick solves almost every problem of the sheet.

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1. (IMC 2014) Consider the sequence

$$(a_n)_{n\geqslant 1} = (1, 1, 2, 1, 2, 3, 1, 2, 3, 4, 1, 2, 3, 4, 5, 1, \dots).$$

Find all pairs (α, β) of positive reals such that $\lim_{n\to\infty} \frac{\sum_{k=1}^{n} a_k}{n^\alpha} = \beta$.

2. (IMC 2000) Let $f\colon \mathbb{R} \to (0,\infty)$ be an increasing differentiable function with $\lim_{x\to\infty} f(x) = \infty$ and f' bounded, and let $F(x) = \int_0^x f(t)\,dt$. Define $a_0 = 1$, $a_{n+1} = a_n + \frac{1}{f(a_n)}$ and $b_n = F^{-1}(n)$. Prove that $\lim_{n\to\infty}(a_n - b_n) = 0$.

3. (VJIMC 2005) Let $(x_n)_{n\geqslant 2}$ be a sequence of real numbers such that $x_2 > 0$ and $x_{n+1} = -1 + \sqrt[n]{1+nx_n}$ for $n \geqslant 2$. Find $\lim_{n\to\infty} x_n$ and $\lim_{n\to\infty} nx_n$.

4. (AKHIMO 2022) Let $x_1 \in \left(0, \frac{\pi}{2}\right)$ and $x_{n+1} = \sqrt[3]{\frac{\sin^3 x_1 + \cdots + \sin^3 x_n}{n}}$ for $n \geqslant 1$. Find $\lim_{n\to\infty} x_n^2 \ln n$.

5. (IMC 2026) Let $x_1 > 0$ and define the sequence (x_n) by $x_{n+1} = \arctan \frac{x_1 + x_2 + \cdots + x_n}{n}$ for all $n \geqslant 1$. Find $\lim_{n\to\infty} x_n \sqrt{\ln n}$.